



Module Presenter's Manual

for

Java Programming - II

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Ver. 1.0

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	April 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Explain Java utility APIs
- Describe Generics feature in Java
- Explain File handling and Stream API
- Learn the concept of threads in Java and explains how to create a Thread object
- Explain Multithreading and Concurrency
- Learn the software API for database connectivity, JDBC
- Explain Design Patterns and Data Structures
- Explain JavaFX, the framework for building modern, interactive, and visually appealing user interfaces for Java applications
- Learn JavaFX Text, Transformation, and Shapes
- Explain JavaFX Layouts, UI, and Charts
- Explain media integration with JavaFX

2. Information on Session Allocation

Module	Online Hours
Java Programming - II	40

Throughout this Presenter's Manual, the module **Java Programming - II** will be referred to as **JavaFX**.

3. Module Deliverables on OnlineVarsity

To aid the teaching process, following are the deliverables:

Resources available on Onlinevarsity:

Feature - Description/Functionality
Ebook -Student reading material in the form of PDF
Download eBook - Student has the option to download the subject related e-book and read offline
Practice 4 Me - Student can test and evaluate their understanding of module related topics.
Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
References - Student can access additional subject related material for reading.

4. Week-wise Session Schedule

- A session is of 2 hours duration.

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 JAVAFX – TL1	Session 2 JAVAFX – TL2	Session 3 JAVAFX – TL3	Session 4 JAVAFX – TL4
2	Session 5 JAVAFX -TL5	Session 6 JAVAFX – TL6	Session 7 JAVAFX – TL7	Session 8 JAVAFX – TL8
3	Session 9 JAVAFX -TL9	Session 10 JAVAFX – TL10	Session 11 JAVAFX – TL11	Session 12 JAVAFX – TL12
4	Session 13 JAVAFX -TL13	Session 14 JAVAFX – TL14	Session 15 JAVAFX – TL15	Session 16 JAVAFX – TL16
5	Session 17 JAVAFX -TL17	Session 18 JAVAFX – TL18	Session 19 JAVAFX – TL19	Session 20 JAVAFX – TL20

JavaFX: Java Programming - II
TL: Online Session

5. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	JAVAFX – TL1	All the topics as listed below from Session 1 and Session 2 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 1 – Java Utility APIs</u> ➤ Explain java.util package ➤ Explain List classes and interfaces ➤ Explain Set classes and interfaces ➤ Explain Map classes and interfaces ➤ Explain Queues and Arrays <u>Session 2 – Generics</u> ➤ Identify the necessity for Generics ➤ List the advantages and limitations of Generics ➤ Explain generic class declaration and instantiation ➤ Define generic methods ➤ Describe working with generic methods ➤ Explain the wildcard argument ➤ Describe the use of inheritance with Generics ➤ Explain type inference	<u>JavaFX and Advanced Java</u> SG - Session 1 & 2 XP - Session 1 & 2 TG - Session 1 & 2
2	JAVAFX – TL2	The Try It Yourself questions of Session 1 and Session 2 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 1 & 2
3	JAVAFX – TL3	All the topics as listed below from Session 3 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 3 – File Handling, Stream API, and Related Concepts</u> ➤ Define data streams ➤ Identify purpose of the File class and its methods ➤ Explain DataInput and DataOutput interfaces ➤ Outline byte stream and character streams in java.io package	<u>JavaFX and Advanced Java</u> SG - Session 3 XP - Session 3 TG - Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain InputStream and OutputStream classes ➤ Describe BufferedInputStream and BufferedOutputStream classes ➤ Identify Character stream classes ➤ Define Serialization and identify the necessity and purpose of Serialization ➤ Elaborate on Console class ➤ Identify Unsafe Operations on foreign memory ➤ Describe uses of foreign functions	
4	JAVAFX – TL4	All the topics as listed below from Session 4 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 4 – Threading</u> ➤ Explain the Thread class ➤ Elaborate on the process of creating threads ➤ Identify Thread states ➤ List methods of Thread class ➤ Explain managing threads ➤ Describe daemon threads ➤ Explain Scoped values in JDK 20 ➤ Describe how to create Virtual Threads ➤ Explain the difference between Virtual threads and OS threads	<u>JavaFX and Advanced Java</u> SG - Session 4 XP - Session 4 TG - Session 4
5	JAVAFX – TL5	The Try It Yourself questions of Session 3 and Session 4 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 3 & 4
6	JAVAFX – TL6	All the topics as listed below from Session 5 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 5 – Multithreading and Concurrency</u> ➤ Define multithreading ➤ Differentiate between multithreading and multitasking ➤ Explain the use of <code>isAlive()</code> and <code>join()</code> method ➤ Explain race conditions and ways to overcome them ➤ Elaborate on intrinsic lock and synchronization	<u>JavaFX and Advanced Java</u> SG - Session 5 XP - Session 5 TG - Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Identify atomic access ➤ Identify the use of wait() and notify() methods ➤ Define deadlock and the ways to overcome deadlock ➤ Explain java.util.concurrent collections ➤ Explain about Structured Concurrency ➤ Elaborate on the process to build radix sort program with Foreign API ➤ Outline the use of off-heap and on-heap memory areas	
7	JAVAFX – TL7	All the topics as listed below from Session 6 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 6 – JDBC API</u> ➤ Explain basics of JDBC ➤ Elaborate on JDBC architecture ➤ Identify and describe different types of processing models ➤ Outline JDBC Driver Types ➤ List the steps involved in JDBC application development ➤ Explain Database Meta Information ➤ Elaborate on parameterized queries in JDBC API	<u>JavaFX and Advanced Java</u> SG - Session 6 XP - Session 6 TG - Session 6
8	JAVAFX – TL8	The Try It Yourself questions of Session 5 and Session 6 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 5 & 6
9	JAVAFX – TL9	All the topics as listed below from Session 7 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 7 – Advanced JDBC</u> ➤ List and describe scrollable result sets ➤ List different types of ResultSet and row-positioning methods ➤ Explain stored procedures ➤ Explain how to call a stored procedure using JDBC API ➤ Identify the steps to update records ➤ Explain the steps of implementing transactions using JDBC	<u>JavaFX and Advanced Java</u> SG - Session 7 XP - Session 7 TG - Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ List the enhancements of JDBC 4.3 API ➤ Explain Rowset and its type ➤ Describe JDBC 4.1 RowSetProvider and RowSetFactory Interfaces	
10	JAVAFX – TL10	All the topics as listed below from Session 8 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 8 – Design Patterns and Other Advanced Features</u> ➤ Explain design patterns ➤ Describe delegation, composition, and aggregation ➤ Describe internationalization and localization ➤ Explain the enhancements of java.util.concurrent package ➤ Describe atomic operations with the new set of classes of the java.util.concurrent.atomic package ➤ Explain the new features of ForkJoinPool	<u>JavaFX and Advanced Java</u> SG - Session 8 XP - Session 8 TG - Session 8
11	JAVAFX – TL11	All the topics as listed below from Session 9 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 9 – Java Data Structures</u> ➤ Explain the Enumeration interface ➤ Describe the BitSet class ➤ Describe the Stack classes ➤ Explain the Dictionary classes	<u>JavaFX and Advanced Java</u> SG - Session 9 XP - Session 9 TG - Session 9
12	JAVAFX – TL12	The Try It Yourself questions of Session 7 to Session 9 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 7-9
13	JAVAFX – TL13	All the topics as listed below from Session 10 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 10 – New and Advanced Features of Java</u> ➤ List the enhancements in switch-case ➤ Explain new FileSystems methods ➤ Explain the not Predicate method ➤ Describe Records and Text Blocks ➤ Explain Flow, Stack Walking, and HTTP Client APIs	<u>JavaFX and Advanced Java</u> SG - Session 10 XP - Session 10 TG - Session 10

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe accounting style currency format support ➤ Explain CompactNumberFormat class ➤ Describe Sealed Classes for encapsulation and enhancing security	
14	JAVAFX – TL14	All the topics as listed below from Session 11 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 11 – Introduction to JavaFX</u> ➤ Describe an overview of JavaFX ➤ Summarize the history of JavaFX ➤ Explain installation process of Eclipse ➤ Elaborate JavaFX Architecture ➤ Explain how to work with JavaFX using Eclipse ➤ Explain JavaFX application structure ➤ Elaborate features of JavaFX	<u>JavaFX and Advanced Java</u> SG - Session 11 XP - Session 11 TG - Session 11
15	JAVAFX – TL15	All the topics as listed below from Session 12 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 12 – JavaFX Text, Transformation, and Shapes</u> ➤ Explain JavaFX Text ➤ Identify JavaFX Text Properties ➤ Explain JavaFX Transformation ➤ Describe JavaFX Translation ➤ Explain JavaFX Rotation ➤ Describe JavaFX Scaling ➤ Explain JavaFX Shearing ➤ Describe JavaFX 2D and 3D Shapes	<u>JavaFX and Advanced Java</u> SG - Session 12 XP - Session 12 TG - Session 12
16	JAVAFX – TL16	The Try It Yourself questions of Session 10 to Session 12 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 10-12
17	JAVAFX – TL17	All the topics as listed below from Session 13 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 13 – JavaFX Layouts, UI, and Charts</u> ➤ Explain JavaFX Layouts ➤ Describe JavaFX CSS ➤ Describe JavaFX BorderPane ➤ Explain JavaFX HBox and VBox	<u>JavaFX and Advanced Java</u> SG - Session 13 XP - Session 13 TG - Session 13

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Describe JavaFX UI Controls ➤ Describe JavaFX UI Properties ➤ Explain JavaFX Charts and their types	
18	JAVAFX – TL18	All the topics as listed below from Session 14 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 14 – JavaFX Event Handling</u> ➤ Explain JavaFX Event Handling ➤ Identify JavaFX Convenience methods ➤ List and describe JavaFX Event Filters ➤ Elaborate on JavaFX Event Handlers	<u>JavaFX and Advanced Java</u> SG - Session 14 XP - Session 14 TG - Session 14
19	JAVAFX – TL19	All the topics as listed below from Session 15 of <i>JavaFX and Advanced Java</i> book should be covered in this session. <u>Session 15 – Media with JavaFX</u> ➤ Illustrate an overview of using media with JavaFX ➤ Explain properties and constructors of Media class ➤ Explain the process to play audio with JavaFX ➤ Explain the process to play video with JavaFX	<u>JavaFX and Advanced Java</u> SG - Session 15 XP - Session 15 TG - Session 15
20	JAVAFX – TL20	The Try It Yourself questions of Session 13 to Session 15 of <i>JavaFX and Advanced Java</i> book should be covered in this session.	<u>JavaFX and Advanced Java</u> Session 13-15

6. Library References

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| ➤ The Well-Grounded Java Developer, Second Edition by Benjamin J. Evans, Jason Clark, and Martijn Verburg |
| ➤ Learn JavaFX 17: Building User Experience and Interfaces with Java by Kishori Sharan and Peter Späth |
| ➤ JAVA: THE COMPLETE REFERENCE ,12TH EDITION by Herbert Schildt |